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Modified Kolmogorov–Smirnov Test Procedures with Application to Arbitrarily Right-Censored Data

Thomas R. Fleming, Judith R. O'Fallon and Peter C. O'Brien

Department of Medical Statistics and Epidemiology, Mayo Clinic, Rochester, Minnesota 55901, U.S.A.

and

David P. Harrington

Department of Applied Mathematics and Computer Science, University of Virginia, Charlottesville, Virginia 22901, U.S.A.

SUMMARY

The classical Kolmogorov one-sample goodness-of-fit and Smirnov two-sample test procedures are modified in an attempt to obtain increased power when applied to uncensored data. Both are then generalized for use with arbitrarily right-censored data. Motivation for the formulation of the generalized procedures is provided heuristically by appealing to the basic concept upon which the procedures for uncensored data are based; a theoretical motivation is provided by asymptotic weak convergence results. The size and power of the generalized Smirnov two-sample procedure are evaluated for small and moderate sample sizes using Monte Carlo simulations. Results of the simulations are also used to make comparisons with the Gehan–Wilcoxon and logrank two-sample test procedures. The generalized Smirnov procedure is found to maintain the size of the test in nearly all the cases studied, although it is conservative in small samples of heavily censored data. It is found to be more powerful than both the Gehan–Wilcoxon and the logrank procedures for the non-Lehmann alternatives considered.

1. Introduction

In this paper we propose a modification of the classical Kolmogorov one-sample goodness-of-fit and Smirnov two-sample test procedures in an attempt to obtain increased power when dealing with uncensored data. These modified procedures are particularly appealing as they allow for straightforward generalization to procedures, relatively easy to employ, which are applicable when the data are subject to random right censorship. Without loss of generality, we shall assume that the data are survival times measured from 'onset of treatment' until 'death'. The generalized procedures, which are of central interest in this paper, differ from those proposed by Dufour and Maag (1978), Barr and Davidson (1973), Koziol and Byar (1975), and Schey (1977), all of whom considered only restrictive forms of censoring.

Although a one-sample goodness-of-fit procedure will be formulated, an evaluation of its properties and a discussion of its relationship to other censored-data goodness-of-fit

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procedures will be presented in a later paper. Here we will concentrate on the formulation of a generalized Smirnov two-sample test procedure and an evaluation of its properties.

Efficient procedures for comparing two samples subject to random right censorship have been proposed previously. The logrank test, proposed by Mantel (1966) and discussed extensively by others [see especially Cox (1972), Breslow (1975), Peto (1972), Peto and Peto (1972) and Peto and Pike (1973)], is known to be a fully efficient rank test under Lehmann alternatives (alternatives in which the relative hazard is constant) when censoring distributions are equal. In general, then, the logrank test tends to be sensitive to distributional differences which, on semilog plots, are most evident late in time. In comparison, a generalized Wilcoxon procedure proposed by Gehan (1965) has been found to be more powerful in detecting differences very evident early in time (Lee, Desu and Gehan, 1975; Prentice and Marek, 1979; also some results by E. T. Lee and D. R. Thomas presented at meetings in Boston in 1976).

The present research was motivated to an extent by our finding that both the logrank test and the Gehan–Wilcoxon test may be insensitive to certain commonly occurring departures from the null hypothesis. Specifically, it has been our frequent experience that substantial differences between two survival distributions may be apparent at one point in time, but fail to exist elsewhere. For example, certain treatments for coronary heart disease yield remarkably improved long-term survival, even though survival immediately following onset of treatment may be worse than that obtained with less aggressive alternative treatments. It is also common to observe that a sample of individuals treated with radiation or chemotherapeutic antitumor regimens experience an excellent improvement in short-term survival but little or no long-term survival advantage when compared to a control population.

Many departures from the null hypothesis of the type described above fall within the class commonly referred to as 'crossing-hazards alternatives.' These alternatives arise when the hazard functions $\nu_i(t) = -(d/dt)\{\ln S_i(t)\}$ of the survival distributions $S_i(t)$, $i = 1, 2$, cross. Careful inspection of the logrank and Gehan–Wilcoxon test statistics suggests that test procedures based upon these statistics may be insensitive to such departures.

The obvious statistic for which a comparison of distributions is based upon the maximal observed difference is, of course, a Smirnov statistic. Thus, this type of statistic is sensitive to any difference in distributions, which is particularly evident at at least one point in time. The Smirnov-type statistic we propose is formulated using Nelson's estimator $\hat{\beta}_i(t)$ (Nelson, 1969) of the cumulative hazard function $\beta_i(t) = \int \nu_i(s)\, ds$ (integration from 0 to t) rather than the empirical survival function used by Smirnov (1939). Operating characteristics of this new procedure, based upon convergence of an underlying process to a time-transformed Brownian bridge, differ considerably from those of a procedure proposed recently by Aalen (1976), which is based upon convergence of an underlying process to a time-transformed Brownian motion. Properties of Aalen's procedure and of others of similar construction are discussed at length by Fleming and Harrington (1981).

Modification of the classical Kolmogorov–Smirnov statistics in uncensored data, as well as their generalization to randomly right-censored data, are derived in §2. An algorithm for computing each of these statistics is also presented in §2, followed by illustrative examples in §3. The asymptotic distributions of all statistics under the null hypothesis are investigated in Appendix 1, with the null hypothesis distributions of the two-sample statistics being further investigated in terms of Monte Carlo simulations in §4. Power comparisons of the two-sample procedure with the classical Smirnov test in uncensored data and with the logrank and Gehan–Wilcoxon tests in censored data are also provided in §4.

2. Generalized Kolmogorov–Smirnov Test Procedures

2.1 *Statistical Model*

Let S represent the true survival distribution of a population under consideration. In the one-sample goodness-of-fit problem we will be interested in a test of the hypothesis $H_0: S(t) = S^0(t)$ for all $t \geq 0$, where $S^0(t)$ represents the hypothesized distribution. Assume that S and S^0 have cumulative hazard functions $\beta(t) = -\ln S(t)$ and $\beta^0(t) = -\ln S^0(t)$.

In the two-sample problem, consider two populations having true survival functions S_i, $i = 1, 2$. In this situation we will be interested in a test of the hypothesis $H_0: S_1(t) = S_2(t) = S(t)$ for all $t \geq 0$, where $S(t)$ is unspecified. For $i = 1, 2$, let $\beta_i(t) = -\ln S_i(t)$. Although only the two-sample notation will now be presented, notation for the one-sample problem can readily be obtained by dropping the subscript i.

To test the null hypothesis H_0, assume that one observes a sample of N_i individuals from population i, for each of whom the survival information is subject to arbitrary random right censorship. Specifically, if the jth individual in sample i has true survival time X_{ij} and censoring time Y_{ij}, then one is able to observe only $Z_{ij} = \min(X_{ij}, Y_{ij})$ and $\delta_{ij} = I_{[X_{ij} \leq Y_{ij}]}$, where $I_{[E]}$ is 1 if event E occurs and 0 otherwise. The survival and censoring distributions for the individuals in sample i are given by $S_i(t) = \mathrm{pr}(X_{ij} \geq t)$ and $C_i(t) = \mathrm{pr}(Y_{ij} \geq t)$, $j = 1, \ldots, N_i$ and $i = 1, 2$. We assume that X_{ij} and Y_{ij} are independent, although a slightly weaker assumption given in Appendix 1 is found to be sufficient.

In summary, in the statistical model of random right censorship under consideration, it is not assumed that the censoring distributions are the same for both samples, but it is required that a relationship exists between censoring and survival distributions, which is nearly as strong as statistical independence.

2.2 *Estimation*

Before the estimators to be employed are defined, it is necessary to establish some additional notation. Again, although only two-sample notation will be discussed, that for the one-sample case may be obtained by dropping the subscript i. Let $\{t_j : j = 1, \ldots, m\}$ be the set of m distinct observation times in our pooled sample; that is, distinct times of death or censorship. Let $\{T_j : j = 1, \ldots, d\}$ be the subset of d distinct death times, and let $\{\tau_j : j = 1, \ldots, c\}$ be the subset of c distinct censorship times. Clearly, $m \leq d + c$. Let $N_i(t)$ represent the number of individuals in sample i under observation at time t, *prior* to deaths or censorships occurring at t. Further, $D_i(t)$ and $L_i(t)$ represent the number of deaths and censorships, respectively, in sample i at time t.

Recall that $\beta_i(t) = -\ln S_i(t)$. Similarly define $\alpha_i(t) = -\ln C_i(t)$. Nelson's cumulative hazard estimators (Nelson, 1969) $\hat{\beta}_i(t)$ and $\hat{\alpha}_i(t)$ are then given by

$$\hat{\beta}_i(t) = \sum_{T_j \leq t} \sum_{k=0}^{D_i(T_j)-1} \{N_i(T_j) - k\}^{-1}$$

and

$$\hat{\alpha}_i(t) = \sum_{\tau_j \leq t} \sum_{k=0}^{L_i(\tau_j)-1} \{N_i(\tau_j) - D_i(\tau_j) - k\}^{-1},$$

where we have adopted the convention of breaking ties between deaths and censorships in sample i by assuming that the deaths occurred infinitesimally earlier, and where we define $\sum_0^{-1} f(k) \equiv 0$ for any f. Finally, define the survival distribution estimator of $S_i(t)$ by $\hat{S}_i(t) = \exp\{-\hat{\beta}_i(t)\}$, and define $\hat{C}_i(t) = \exp\{-\hat{\alpha}_i(t)\}$.

2.3 *Modification of Classical Kolmogorov–Smirnov Statistics in Uncensored Data*

The following results provide the theoretical motivation for the modification of the Kolmogorov goodness-of-fit and Smirnov two-sample test procedures in uncensored data. Later, in §2.4 and §2.5, formulas are provided for the calculation of these modified statistics. To give, whenever possible, a unified presentation of the one- and two-sample situations, let the ordered pair $\{S_l(t), S_u(t)\}$ represent both the one-sample survival distributions $\{S^0(t), S(t)\}$ as well as the two-sample survival distributions $\{S_1(t), S_2(t)\}$. The cumulative hazard functions for $\{S_l(t), S_u(t)\}$ are denoted by $\{\beta_l(t), \beta_u(t)\}$.

We begin by briefly deriving a result which will lead to the conjecture that a modification of the classical Kolmogorov–Smirnov statistics will yield increased power in uncensored data. Observe that

$$\begin{aligned} S_u(t)-S_l(t) &= [\exp\{-\beta_l(t)\}][\exp\{\beta_l(t)-\beta_u(t)\}-1] \\ &= \exp\{-\beta_l(t)\}\int_0^t \exp\{\beta_l(s)-\beta_u(s)\}\, d\{\beta_l(s)-\beta_u(s)\} \\ &= \int_0^t \{S_u(s)S_l(t)/S_l(s)\}\, d\{\beta_l(s)-\beta_u(s)\}. \end{aligned} \tag{2.1}$$

If $S_l(s) \doteq S_u(s)$ for all $s \leq t$, it follows that $S_u(s)S_l(t)/S_l(s) \doteq \frac{1}{2}\{S_l(t)+S_u(t)\}$, and from (2.1) it follows that

$$S_u(t)-S_l(t) \doteq \tfrac{1}{2}\{S_l(t)+S_u(t)\}\{\beta_l(t)-\beta_u(t)\}. \tag{2.2}$$

When the null hypothesis, $H_0: S_l(t)=S_u(t)$ for $t \geq 0$, holds, both sides in (2.2) are zero. However, when the null hypothesis fails to hold, the following interesting relationship exists and is proved in Appendix 2 using elementary techniques of calculus:

Lemma 1. When $S_l(t) \neq S_u(t)$,

$$\tfrac{1}{2}\{S_l(t)+S_u(t)\}\{\beta_l(t)-\beta_u(t)\}/\{S_u(t)-S_l(t)\} > 1.$$

This ratio differs from 1 to any noticeable extent only when either $S_l(t)$ or $S_u(t)$ is very near zero.

Formulation of the classical Kolmogorov–Smirnov test statistics is based upon the estimator of the difference $S_u(t)-S_l(t)$. Consider briefly the one-sample case with sample size N: if we let $\tilde{S}(t)$ represent the empirical distribution function estimator of $S(t)$, then the classical one-sided Kolmogorov test statistic is

$$\mathscr{X}_N \equiv \sup_{0\leq t\leq t_m} N^{\frac{1}{2}}\{\tilde{S}(t)-S^0(t)\}.$$

Observe that, for $t \leq t_m$,

$$|\tilde{S}(t)-S^0(t)| \doteq |\hat{S}(t)-S^0(t)| \leq \tfrac{1}{2}\{\hat{S}(t)+S^0(t)\}\, |\beta^0(t)-\hat{\beta}(t)|, \tag{2.3}$$

where the inequality of (2.3) follows directly from Lemma 1, and the approximation in (2.3) is excellent, at least in moderate to large sample sizes.

In view of (2.3), it is of interest to compare the behavior of $\mathscr{X}_N$ with that of the modified one-sided Kolmogorov-type test statistic

$$\mathscr{Y}_N(t_m) \equiv \sup_{0\leq t\leq t_m} \tfrac{1}{2}\{\hat{S}(t)+S^0(t)\}\int_0^t N^{\frac{1}{2}} I_{[N(s)>0]}\, d\{\beta^0(s)-\hat{\beta}(s)\}.$$

Let τ be a fixed positive number satisfying $\text{pr}(Z_{ij} \geqslant \tau) > 0$ for all i, j. It follows from Theorem A.1 in Appendix 1 that when the null hypothesis is true the statistic $\mathcal{Y}_N(\tau)$ has an asymptotic distribution which is stochastically smaller than the asymptotic distribution of the classical Kolmogorov statistic $\mathcal{X}_N$, with equality of the two limiting distributions occurring as $S(\tau) \to 0$. Suppose that the small sample distribution of the statistic $\mathcal{X}_N$ is approximated by its asymptotic distribution, and that of the statistic $\mathcal{Y}_N(t_m)$ is approximated by using the asymptotic distribution of $\mathcal{Y}_N(\tau)$ and then setting $\tau = t_m$. The stochastic ordering of these asymptotic distributions, together with the result in (2.3), leads to the conjecture that one would obtain greater power using the test statistic $\mathcal{Y}_N(t_m)$ rather than $\mathcal{X}_N$, particularly against departures from H_0, which are most evident later in time. This same reasoning also leads to the suspicion that, when H_0 is true, use of $\mathcal{Y}_N(t_m)$ will result in rejections of the null hypothesis more frequently than use of $\mathcal{X}_N$. However, since $\mathcal{X}_N$ is known to be quite conservative, particularly in small samples, one might conjecture that $\mathcal{Y}_N(t_m)$ will be less conservative while still preserving the size of the test. Although the discussion up to this point has been in relation to the one-sided one-sample test, clearly the same reasoning applies to the two-sided test and to the two-sample situation.

Results of simulations presented in §4.1 confirm these conjectures in the two-sample situation. In small and moderate sample sizes the two-sample procedure analogous to that based upon $\mathcal{Y}_N(t_m)$ does preserve the size of the test when the null hypothesis is true and, under the various alternatives to H_0 which we inspected, has as good or better power than the classical two-sample Smirnov procedure. In large samples we can expect from the asymptotic distribution results that both of these two-sample procedures should essentially preserve the size of the test.

In conclusion, it appears that one- and two-sample test procedures based upon estimation of the difference $\frac{1}{2}\{S_l(t)+S_u(t)\}\{\beta_l(t)-\beta_u(t)\}$ will behave at least as well as, if not better than, the classical uncensored data procedures based upon $\{S_u(t)-S_l(t)\}$. Furthermore, important additional motivation for the uncensored data modification of the classical Kolmogorov–Smirnov test procedures arises from the fact that these modified procedures allow easy generalization to censored data, as will be seen in the next section, with the generalized procedures requiring relatively straightforward computation.

2.4 *Generalization of Modified Kolmogorov Goodness-of-Fit Test Procedures to Arbitrarily Right-Censored Data*

For uncensored data it has been proposed that a one-sample goodness-of-fit test procedure could be based upon inspection of the difference

$$\tfrac{1}{2}\{\hat{S}(t)+S^0(t)\}\int_0^t N^{\frac{1}{2}} I_{[N(s)>0]}\, d\{\beta^0(s)-\hat{\beta}(s)\}.$$

Clearly, changes in the cumulative hazard functions are weighted by the square root of the sample size.

Effectively, in censored data the amount of information available to estimate the change in the survival or cumulative hazard function at time s is only a fraction $C(s)$ of that available in uncensored data. Hence our goodness-of-fit test procedures in censored data will be based upon inspection of the difference

$$Y_N(t) = \tfrac{1}{2}\{\hat{S}(t)+S^0(t)\}\int_0^t \{N\hat{C}(s^-)\}^{\frac{1}{2}} I_{[N(s)>0]}\, d\{\beta^0(s)-\hat{\beta}(s)\}$$

at every t, where $f(t^-)$ denotes the left-hand limit of the function f at t. The left-hand limit of the estimator of $C(s)$ is used for consistency with the convention employed in breaking ties between deaths and censorships.

Recall that t_m is the largest observation time in the sample. Usually, the generalized one-sided Kolmogorov goodness-of-fit test statistic is given by

$$\mathcal{Y}_N(t_m) = \sup_{0 \leq t \leq t_m} Y_N(t).$$

Effectively, then, one is employing the statistic $\mathcal{Y}_N(t_m)$ to test the hypothesis $H_0: S(t) = S^0(t)$ for $0 \leq t \leq t_m$. However, in some applications the data are heavily censored beyond a time τ; in such situations one may have very good power to detect substantial deviations of $S(t)$ from $S^0(t)$ over the interval $[0, \tau]$, but will have relatively little sensitivity using $\mathcal{Y}_N(t_m)$ to detect substantial deviations of $S(t)$ from $S^0(t)$, which arise after τ. Therefore, if particularly heavy censorship beyond some time τ is expected, one has the capability only to test $H_0': S(t) = S^0(t)$ for $0 \leq t \leq T'$, where $T' \equiv \min(t_m, \tau)$. The statistic $\mathcal{Y}_N(T') = \sup Y_N(t)$ (supremum over $[0, T']$) would then be the appropriate means of testing H_0'.

We will now define the procedure to calculate the generalized one- and two-sided Kolmogorov goodness-of-fit tests based upon the statistics

$$\mathcal{Y}_N(t_m) \equiv \sup_{0 \leq t \leq t_m} Y_N(t) \quad \text{and} \quad \tilde{\mathcal{Y}}_N(t_m) \equiv \sup_{0 \leq t \leq t_m} |Y_N(t)|:$$

(i) Set $\hat{\alpha}(t_0) = \hat{\beta}(t_0) = 0$ and recursively calculate, for $j = 1, \ldots, m$,

$$\hat{\beta}(t_j) = \hat{\beta}(t_{j-1}) + \sum_{k=0}^{D(t_j)-1} \{N(t_j) - k\}^{-1}$$

and

$$\hat{\alpha}(t_j) = \hat{\alpha}(t_{j-1}) + \sum_{k=0}^{L(t_j)-1} \{N(t_j) - D(t_j) - k\}^{-1}.$$

Setting $A(t_0) = B(t_0) = 0$, recursively calculate, for $j = 1, \ldots, m$,

$$A(t_j) = A(t_{j-1}) + \exp\{-\tfrac{1}{2}\hat{\alpha}(t_{j-1})\}[\ln\{S^0(t_{j-1})/S^0(t_j)\}]$$

and

$$B(t_j) = B(t_{j-1}) + \exp\{-\tfrac{1}{2}\hat{\alpha}(t_{j-1})\}\left[\sum_{k=0}^{D(t_j)-1} \{N(t_j) - k\}^{-1}\right].$$

(ii) For $j = m$ and for all j such that $t_j \in \{T_1, \ldots, T_d\}$, calculate

$$Y_N(t_j^-) = \tfrac{1}{2}N^{\frac{1}{2}}[\exp\{-\hat{\beta}(t_{j-1})\} + S^0(t_j)]\{A(t_j) - B(t_{j-1})\}$$

and

$$Y_N(t_j) = \tfrac{1}{2}N^{\frac{1}{2}}[\exp\{-\hat{\beta}(t_j)\} + S^0(t_j)]\{A(t_j) - B(t_j)\}.$$

(iii) Set $V = \max\{0, Y_N(T_j^-), Y_N(T_j), Y_N(t_m): j = 1, \ldots, d\}$ and

$$A = \max\{|Y_N(T_j^-)|, |Y_N(T_j)|, |Y_N(t_m)|: j = 1, \ldots, d\}.$$

(iv) (a) Set $R = 1 - \frac{1}{2}[\exp\{-\hat{\beta}(t_m)\} + S^0(t_m)]$.

(b) Calculate the P-value for the one-sided test (see Appendix 1):

$$P = p(V, R) = 1 - \Phi\{V/(R - R^2)^{\frac{1}{2}}\} + \Phi\{V(2R - 1)/(R - R^2)^{\frac{1}{2}}\} \exp(-2V^2), \quad (2.4)$$

where $\Phi(x)$ is the cumulative distribution function of the standard normal distribution evaluated at x. Alternatively:

(c) Calculate the approximate P-value for the two-sided test: $P \simeq 2p(A, R)$ when $p(A, R) < .40$ (see Appendix 1).

Note that if $S^0(t_m) = 0$, the null hypothesis is rejected since an event in the sample has occurred which is outside the support of the distribution under the null hypothesis.

2.5 *Generalization of the Modified Smirnov Two-Sample Test Procedure to Arbitrarily Right-Censored Data*

It is apparent that the two-sample uncensored data modification of the classical Smirnov procedure, which is analogous to that just proposed for one sample, would be based upon inspection of the difference

$$\tfrac{1}{2}\{\hat{S}_1(t) + \hat{S}_2(t)\} \int_0^t \left(\frac{N_1 N_2}{N_1 + N_2}\right)^{\frac{1}{2}} I_{[N_1(s)N_2(s)>0]}\, d\{\hat{\beta}_1(s) - \hat{\beta}_2(s)\}.$$

For reasons identical to those cited when generalizing the one-sample test to censored data, we will base our two-sample test procedure in censored data upon inspection of the difference

$$Y_{N_1,N_2}(t) = \tfrac{1}{2}\{\hat{S}_1(t) + \hat{S}_2(t)\} \int_0^t \left\{\frac{N_1 \hat{C}_1(s^-) N_2 \hat{C}_2(s^-)}{N_1 \hat{C}_1(s^-) + N_2 \hat{C}_2(s^-)}\right\}^{\frac{1}{2}} I_{[N_1(s)N_2(s)>0]}\, d\{\hat{\beta}_1(s) - \hat{\beta}_2(s)\}$$

at every t.

Define $T = \max\{t_j : N_1(t_j) N_2(t_j) > 0\}$. Usually the generalized one-sided Smirnov two-sample test statistic, given by

$$\mathscr{Y}_{N_1,N_2}(T) = \sup_{0 \leqslant t \leqslant T} Y_{N_1,N_2}(t),$$

would be employed to test the hypothesis $H_0 : S_1(t) = S_2(t)$ for $0 \leqslant t \leqslant T$. However, as discussed in §2.4, suppose that one expects the data to be heavily censored in one or both samples beyond a time τ; it would then be more appropriate simply to attempt to test $H_0' : S_1(t) = S_2(t)$ for $0 \leqslant t \leqslant T'$, by generating the statistic $\mathscr{Y}_{N_1,N_2}(T')$, where $T' = \min(T, \tau)$. (When one selects a τ such that $\mathrm{pr}(Z_{ij} > \tau) > 0$, the asymptotic distribution results employed in Step (v) of the following algorithm apply directly.)

We will now define the procedure to calculate the generalized one- and two-sided Smirnov two-sample tests based upon the statistics

$$\mathscr{Y}_{N_1,N_2}(T) \equiv \sup_{0 \leqslant t \leqslant T} Y_{N_1,N_2}(t) \quad \text{and} \quad \tilde{\mathscr{Y}}_{N_1,N_2}(T) \equiv \sup_{0 \leqslant t \leqslant T} |Y_{N_1,N_2}(t)|:$$

(to calculate $\mathscr{Y}_{N_1,N_2}(T')$ rather than $\mathscr{Y}_{N_1,N_2}(T)$, simply replace T by T' in Step (i) of the following algorithm).

(i) Define $J = \max\{j : T_j \leqslant T\}$ and recall that $\sum_0^{-1} f(k) \equiv 0$ for any f. For $i = 1, 2$, set $\hat{\beta}_i(T_0) = 0$ and recursively calculate, for all $j = 1, \ldots, J$,

$$\hat{\alpha}_i(T_j^-) = \sum_{k=0}^{N_i - N_i(T_j) - 1} (N_i - k)^{-1} - \hat{\beta}_i(T_{i-1})$$

and

$$\hat{\beta}_i(T_j) = \hat{\beta}_i(T_{j-1}) + \sum_{k=0}^{D_i(T_j) - 1} \{N_i(T_j) - k\}^{-1}.$$

(ii) For all $j=1,\ldots,J$, calculate

$$\eta(T_j)=([N_1\exp\{-\hat{\alpha}_1(T_j^-)\}]^{-1}+[N_2\exp\{-\hat{\alpha}_2(T_j^-)\}]^{-1})^{-\frac{1}{2}}$$

(iii) Set $U(T_0)=0$ and recursively calculate, for all $j\in\{1,\ldots,J\}$,

$$U(T_j)=U(T_{j-1})+\{\eta(T_j)\}\left[\sum_{k=0}^{D_1(T_j)-1}\{N_1(T_j)-k\}^{-1}-\sum_{k=0}^{D_2(T_j)-1}\{N_2(T_j)-k\}^{-1}\right]$$

and

$$Y_{N_1,N_2}(T_j)=\tfrac{1}{2}[\exp\{-\hat{\beta}_1(T_j)\}+\exp\{-\hat{\beta}_2(T_j)\}]U(T_j).$$

(iv) Set $V=\max\{0,\ Y_{N_1,N_2}(T_j): j=1,\ldots,J\}$ and $A=\max\{|Y_{N_1,N_2}(T_j)|: j=1,\ldots,J\}$.
(v) (a) Set $R=1-\frac{1}{2}[\exp\{-\hat{\beta}_1(T_J)\}+\exp\{-\hat{\beta}_2(T_J)\}]$.
 (b) The P-value for the one-sided test is $P=p(V,R)$[see (2.4)].
 (c) The approximate P-value for the two-sided test is $P\simeq 2p(A,R)$ when $p(A,R)\leqslant .40$.

This approximation for the two-sided P-value, given by doubling the value $p(A,R)$, was investigated by Schey (1977). He found the approximation to be acceptable when $2p(A,R)<.80$, and to be excellent when $2p(A,R)<.20$. If one chooses not to use this approximation, Koziol and Byar (1975) have tabulated the two-sided P-values which correspond to each pair (A,R).

In general, calculation of either a one-sided or two-sided P-value requires the evaluation of the formula for $p(y,R)$ given in (2.4). However, when $R\geqslant .75$, $\exp(-2y^2)$ is an excellent approximation to $p(y,R)$, which will lead to slightly conservative estimates. Observe that $p(1.2,.75)=.0543\simeq .0561=\exp\{-2(1.2)^2\}$, and $p(1.5,.75)=.0109\simeq .0111=\exp\{-2(1.5)^2\}$.

3. Examples

In this section, examples are presented which show the calculation of the generalized Smirnov statistic in sets of real data.

Example 1. In a recent study, performed at the Mayo Clinic, of patients having limited Stage II or IIIA ovarian carcinoma, a main goal was to determine whether or not grade of disease was associated with time to progression of disease. At the time of the final analysis of the study, the observations of time to progression of disease (with t^+ denoting a censored observation at time t), for patients with low-grade or well-differentiated cancer, were 28, 89, 175, 195, 309, 377^+, 393^+, 421^+, 447^+, 462, 709^+, 744^+, 770^+, 1106^+ and 1206^+ days. Observation times for patients with high-grade or undifferentiated cancer were 34, 88, 137, 199, 280, 291, 299^+, 300^+, 309, 351, 358, 369, 369, 370, 375, 382, 392, 429^+, 451 and 1119^+ days. Estimates of both distributions are presented in Fig. 1a, with computations needed to evaluate the generalized Smirnov statistic displayed in Table 1.

From the table, $A=1.834$. Since $R=1-\frac{1}{2}[\exp\{-\hat{\beta}_1(462)\}+\exp\{-\hat{\beta}_2(462)\}]=.653$, the P-value for the two-sided test of the association of tumor grade with time to disease progression is given by

$$P=2\left[1-\Phi\left\{\frac{A}{(R-R^2)^{\frac{1}{2}}}\right\}+\Phi\left\{\frac{A(2R-1)}{(R-R^2)^{\frac{1}{2}}}\right\}\exp(-2A^2)\right]$$
$$=2[1-(.99994)+(.881)(.00120)]=.00223.$$

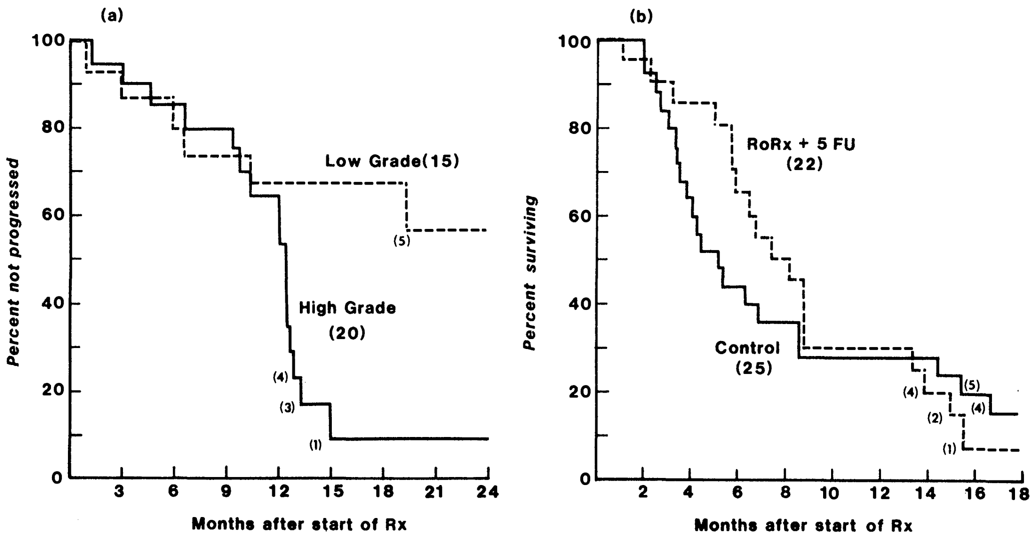

Figure 1. (a) Estimated distributions of time to progression of disease for patients with low-grade and high-grade ovarian carcinoma. (b) Estimated survival distributions for bile duct cancer patients: control population vs patients receiving RöRx + 5-FU.

Table 1

Calculation of the statistics needed to evaluate the two-sided generalized Smirnov statistic in the ovarian carcinoma data

	High-grade				Low-grade						
T_j	$N_1(T_j)$	$D_1(T_j)$	$\hat{\beta}_1(T_j)$	$\hat{\alpha}_1(T_j^-)$	$N_2(T_j)$	$D_2(T_j)$	$\hat{\beta}_2(T_j)$	$\hat{\alpha}_2(T_j^-)$	$\eta(T_j)$	$U(T_j)$	$Y_{N_1,N_2}(T_j)$
28	20	0	0.000	0.000	15	1	0.067	0.000	2.928	−0.195	−0.189
34	20	1	0.050	0.000	14	0	0.067	0.000	2.928	−0.049	−0.046
88	19	1	0.103	0.000	14	0	0.067	0.000	2.928	0.105	0.097
89	18	0	0.103	0.000	14	1	0.138	0.000	2.928	−0.104	−0.092
137	18	1	0.158	0.000	13	0	0.138	0.000	2.928	0.059	0.051
175	17	0	0.158	0.000	13	1	0.215	0.000	2.928	−0.166	−0.138
195	17	0	0.158	0.000	12	1	0.298	0.000	2.928	−0.410	−0.327
199	17	1	0.217	0.000	11	0	0.298	0.000	2.928	−0.238	−0.184
280	16	1	0.280	0.000	11	0	0.298	0.000	2.928	−0.055	−0.041
291	15	1	0.346	0.000	11	0	0.298	0.000	2.928	0.140	0.101
309	12	1	0.430	0.148	11	1	0.389	0.000	2.832	0.119	0.079
351	11	1	0.520	0.148	10	0	0.389	0.000	2.832	0.376	0.239
358	10	1	0.620	0.148	10	0	0.389	0.000	2.832	0.659	0.401
369	9	2	0.857	0.148	10	0	0.389	0.000	2.832	1.328	0.732
370	7	1	0.999	0.148	10	0	0.389	0.000	2.832	1.733	0.906
375	6	1	1.166	0.148	10	0	0.389	0.000	2.832	2.205	1.090
382	5	1	1.366	0.148	9	0	0.389	0.100	2.756	2.756	1.285
392	4	1	1.616	0.148	9	0	0.389	0.100	2.756	3.445	1.509
451	2	1	2.116	0.482	6	0	0.389	0.479	2.303	4.596	1.834
462	1	0	2.116	0.482	6	1	0.556	0.479	2.303	4.212	1.462

Even though R is considerably less than 1, the adjustment for early termination is negligible, i.e. the resulting P-value is nearly equal to that achieved using the formula $P=2\exp(-2A^2)$.

The two-sided logrank procedure, in general quite sensitive to alternatives to the null hypothesis in which substantial differences exist later in time, yielded a P-value of .023, clearly larger than that from the generalized Smirnov procedure. As expected, the two-sided Gehan–Wilcoxon test procedure is somewhat insensitive to this type of alternative to the null hypothesis and yielded a P-value of .134.

Calculation of the generalized Smirnov statistic can be somewhat simplified, particularly in uncensored or lightly censored data, by making the following observation. If the ordered data from the pooled samples contain a string of uncensored observations from one of the samples, one can group these observations, treating them as if they occurred at the same time, without changing the value of the generalized Smirnov test statistic. Remember in so doing that ties between censored and uncensored observations are broken by assuming that the uncensored observations occurred first. The next example shows how these calculations can be done.

Example 2. In another recent Mayo Clinic study, patients with bile duct cancer were followed to determine whether those treated with a combination of radiation treatment (RöRx) and 5-Fluorouracil (5-FU) would survive significantly longer than a control population. Survival times for the RöRx + 5-FU sample were 30, 67, 79^+, 82^+, 95, 148, 170, 171, 176, 193, 200, 221, 243, 261, 262, 263, 399, 414, 446, 446^+, 464 and 777 days, and for the control sample were 57, 58, 74, 79, 89, 98, 101, 104, 110, 118, 125,

Table 2

Calculation of the statistics needed to evaluate the one-sided generalized Smirnov test statistic in the bile duct cancer data

	Control		RöRx + 5-Fu				
j	$N_1(j)$	$D_1(j)$	$N_2(j)$	$D_2(j)$	$\eta(j)$	$U(j)$	$Y_{N_1,N_2}(j)$
1	25	0	22	1	3.421	−0.155	−0.152
2	25	2	21	0	3.421	0.124	0.116
3	23	0	21	1	3.421	−0.039	−0.036
4	23	2	20	0	3.421	0.265	0.233
5	21	1	18	0	3.327	0.424	0.363
6	20	0	18	1	3.327	0.239	0.199
7	20	7	17	0	3.327	1.628	1.133
8	13	0	17	1	3.327	1.432	0.961
9	13	2	16	0	3.327	1.965	1.242
10	11	0	16	3	3.327	1.298	0.724
11	11	1	13	0	3.327	1.601	0.862
12	10	0	13	2	3.327	1.067	0.522
13	10	1	11	0	3.327	1.400	0.657
14	9	0	11	2	3.327	0.765	0.322
15	9	2	9	0	3.327	1.550	0.591
16	7	0	9	5	3.327	−0.930	−0.240
17	7	1	4	0	3.327	−0.455	−0.108
18	6	0	4	1	3.327	−1.286	−0.275
19	6	1	2	0	3.011	−0.785	−0.153
20	5	0	2	1	3.011	−2.290	−0.367
21	5	3	1	0	3.011	0.069	0.007
22	2	0	1	1	3.011	−2.942	−0.202

132, 154, 159, 188, 203, 257, 257, 431, 461, 497, 723, 747, 1313 and 2636 days. Estimators of both distributions are presented in Fig. 1b, with the simplified method of computation of the generalized Smirnov statistic displayed in Table 2.

From the table, $V = 1.242$. Since $R = .931$, the P-value adjustment for early termination is negligible, as is almost always the case when dealing with uncensored or lightly censored data. Thus the one-sided generalized Smirnov P-value is simply given by $P = \exp(-2V^2)$, so that $P = .046$. Gehan–Wilcoxon and logrank one-sided test procedures were also applied to the bile duct cancer data to test whether patients treated with RöRx + 5-FU would survive longer than control patients. In contrast to the one-sided generalized Smirnov procedure which yielded significance at the .05 level, Gehan–Wilcoxon and logrank one-sided test procedures yielded P-values of .127 and .418, respectively.

4. Monte Carlo Simulations

The distribution of the generalized Smirnov two-sample test procedure has been approximated in this paper through the use of asymptotic results. From a practical viewpoint, therefore, it is of interest to consider the size and power in finite samples. In this section we investigate both properties using Monte Carlo simulations.

To evaluate the size of the one-sided generalized Smirnov test procedure, Monte Carlo simulations were performed under various censoring and survival configurations in which the null hypothesis of equality of survival distributions was true. Simulations of various configurations in which the null hypothesis failed to hold were then performed for the purpose of comparing, in these special cases, the power of the generalized Smirnov procedure with that of the Gehan–Wilcoxon and logrank procedures. When uncensored data were generated, comparisons were also made with the classical Smirnov procedure. Because the generalization of the classical Smirnov procedure to arbitrarily right-censored data was motivated by the expectation that such a procedure would be particularly sensitive to substantial differences between survival distributions, which are apparent at one point in time but fail to exist elsewhere, various configurations of this type were chosen for study.

The survival distributions considered were Weibull and piecewise exponential. The censoring was taken to be uniformly distributed over $[0, 1]$ or $[0, 2]$. Since the main purpose was to investigate in small and moderate samples the performance of a procedure derived using asymptotic properties, sample sizes of $N_1 = N_2 = 20$ and $N_1 = N_2 = 50$ were considered.

One thousand pairs of samples (two thousand pairs of samples for evaluation of size) were generated for each selected configuration of survival and censoring distributions for the two populations, and the proportions of samples in which the generalized Smirnov one-sided test rejected H_0 at the $\alpha = .01$ and $\alpha = .05$ significance levels were calculated for each configuration. In order to permit direct comparisons, all procedures being compared were applied to the same data. Simulations were performed by transforming uniform random variables which were generated using the linear congruential method (Knuth, 1969).

4.1 *Uncensored Data*

Results of the Monte Carlo study to evaluate the size and power in uncensored data of the modified Smirnov two-sample test procedure are presented in Table 3. Unlike the classical Smirnov test procedure which is known to be conservative in small samples, a fact

Table 3

Size and power in uncensored data. Monte Carlo estimates of the size and power of the modified Smirnov, classical Smirnov, Wilcoxon, and logrank one-sided test procedures of $H_0: S_1 = S_2$ vs $H_1: S_1 < S_2$ in uncensored data

			Level of test							
			Modified Smirnov		Classical Smirnov		Classical Wilcoxon		Uncensored logrank	
$S_1(t)$	$S_2(t)$	$N_1 = N_2$	.01	.05	.01	.05	.01	.05	.01	.05
Size: 2000 simulations*										
$\lambda_1 = 1$†,	$\lambda_2 = 1$: $t \in (0, \infty)$	20	.0157	.0525	.0069	.0414	.010	.048	.010	.050
		50	.0109	.0408	.0055	.0325	.010	.042	.010	.044
Power: 1000 simulations										
Fig. 2a. 'Proportional hazards'										
$\lambda_1 = 2$,	$\lambda_2 = 1$: $t \in (0, \infty)$	20	.340	.577	.177	.473	.286	.563	.358	.639
		50	.713	.878	.559	.837	.738	.917	.848	.966
Fig. 2b. 'Late difference'										
$W(2, 1)$‡	$W(1, .5)$	20	.344	.525	.095	.301	.061	.174	.173	.449
		50	.810	.909	.393	.724	.104	.297	.603	.862
Fig. 2c. 'Late difference'										
$\lambda_1 = 1$,	$\lambda_2 = 1$: $t \in (0, .8)$	20	.586	.741	.168	.440	.083	.234	.229	.564
$\lambda_1 = 2$,	$\lambda_2 = .2$: $t \in (.8, \infty)$	50	.974	.995	.684	.914	.200	.426	.708	.930
Fig. 2d. 'Middle difference'										
$\lambda_1 = 2$,	$\lambda_2 = 2$: $t \in (0, .1)$	20	.219	.426	.188	.425	.088	.245	.049	.154
$\lambda_1 = 3$,	$\lambda_2 = .75$: $t \in (.1, .4)$	50	.701	.834	.633	.833	.215	.475	.082	.223
$\lambda_1 = .75$,	$\lambda_2 = 3$: $t \in (.4, .7)$									
$\lambda_1 = 1$,	$\lambda_2 = 1$: $t \in (.7, \infty)$									
Fig. 2e. 'Early difference'										
$\lambda_1 = 3$,	$\lambda_2 = .75$: $t \in (0, .2)$	20	.151	.422	.149	.416	.088	.236	.027	.109
$\lambda_1 = .75$,	$\lambda_2 = 3$: $t \in (.2, .4)$	50	.582	.818	.582	.818	.208	.439	.048	.168
$\lambda_1 = 1$,	$\lambda_2 = 1$: $t \in (.4, \infty)$									

* Modified Smirnov and classical Smirnov estimates based upon 10 000 simulations.
† λ_i denotes the constant hazard function of S_i, a piecewise exponential distribution.
‡ $W(\lambda, \alpha)$ denotes a Weibull distribution having survival function $S_i(t) = \exp\{-(\lambda t)^\alpha\}$.

confirmed again in these simulations, the modified Smirnov test procedure has size nearer to the nominal .01 and .05 levels.

In evaluating the power of the newly proposed two-sample test procedure, we inspected several configurations of survival distributions which are shown in Fig. 2. It is apparent in Table 3 that the modified and classical Smirnov one-sided test procedures have nearly identical power against 'early' or 'middle' survival differences which disappear later in time (see Figs 2e and 2d, respectively). On the other hand, if large survival differences are most evident later in time, as in Figs 2a–c, the modified Smirnov procedure has much greater power to detect these differences than does the classical Smirnov procedure.

One configuration having survival differences most evident later in time is the proportional hazards or Lehmann alternative with a doubling in median survival (see Fig. 2a). Since the logrank test is known to be a fully efficient rank test under Lehmann alternatives, one would have anticipated our finding that the logrank test is somewhat more powerful than the modified Smirnov test to this type of alternative. However, the

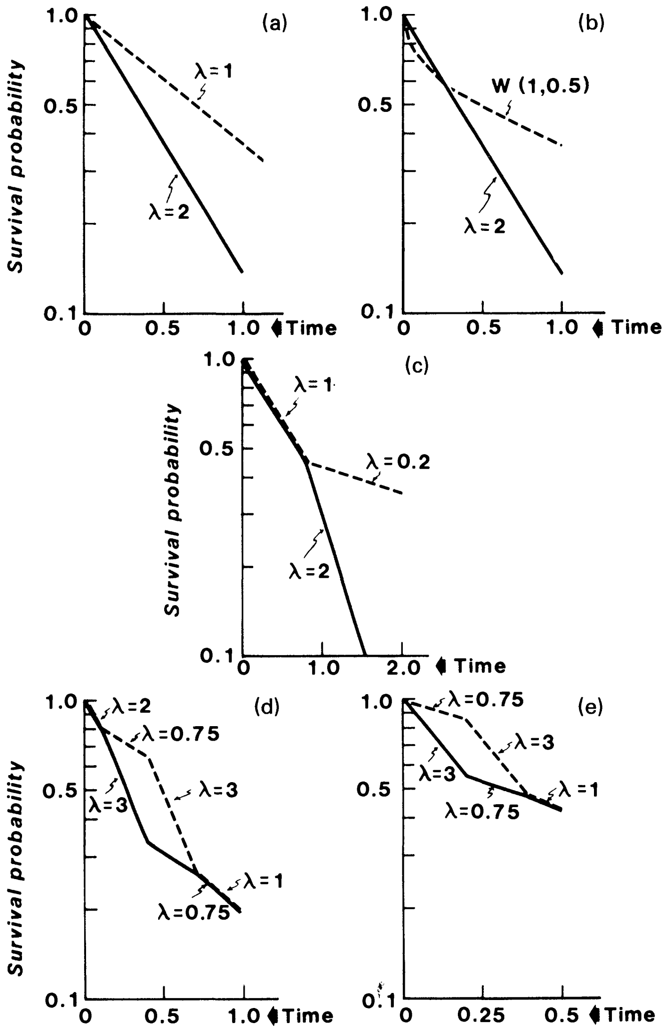

Figure 2. Semilog plots of survival distributions simulated for evaluation of power in uncensored data. Survival distributions are either piecewise exponentials with constant hazard λ over intervals or are Weibull distributions $W(\lambda, \alpha)$ having survival function $S(t) = \exp\{-(\lambda t)^{\alpha}\}$.

difference is not large. Conversely, we found the modified Smirnov test procedure to be more powerful than the Wilcoxon and logrank procedures against all other survival configurations which we considered and where substantial differences between survival distributions were apparent at one point in time but did not appear elsewhere. In some instances the difference in power was quite marked. These results indicate that a generalization of this modified Smirnov test procedure would be very useful in censored data.

4.2 *Censored Data*

Results of Monte Carlo simulations to evaluate the size of the generalized Smirnov test procedure in censored data are presented in Table 4. Various survival and censoring distributions were selected which would allow the evaluation of the generalized Smirnov

Table 4

Size: Monte Carlo estimates of the sizes of the generalized Smirnov, Gehan–Wilcoxon and logrank one-sided test procedures of $H_0: S_1 = S_2$ vs $H_1: S_1 < S_2$ (2000 simulations)

				Level of test					
				Generalized Smirnov		Gehan–Wilcoxon		Logrank	
$S_1(t) = S_2(t)$	$C_1(t) = C_2(t)$	Expected % censored	$N_1 = N_2$	.01	.05	.01	.05	.01	.05
$\lambda_i = .5$*	$U(0, 1)$†	79	20	.007	.030	.009	.063	.007	.052
			50	.006	.031	.007	.045	.007	.049
	$U(0, 2)$	63	20	.007	.030	.007	.044	.005	.040
			50	.009	.041	.008	.051	.010	.055
$\lambda_i = 2$	$U(0, 1)$	43	20	.010	.045	.007	.045	.008	.044
			50	.013	.044	.009	.047	.010	.050
	$U(0, 2)$	25	20	.016	.051	.008	.046	.008	.049
			50	.008	.045	.008	.047	.009	.038
	$U(0, 10^9)$	0	20	.014	.048	.010	.048	.010	.050
			50	.012	.042	.010	.042	.010	.044

* λ_i denotes the constant hazard function of an exponential distribution S_i.
† $U(0, k)$ denotes a uniform distribution over the interval $(0, k)$.

Table 5

Power: Monte Carlo estimates of the power of the Smirnov, Gehan–Wilcoxon, and logrank one-sided test procedures of $H_0: S_1 = S_2$ vs $H_1: S_1 < S_2$ (1000 simulations)

				Level of test					
				Generalized Smirnov		Gehan–Wilcoxon		Logrank	
$S_1(t)$	$S_2(t)$	$C_1(t) = C_2(t)$	$N_1 = N_2$	.01	.05	.01	.05	.01	.05
Fig. 3a. 'Proportional hazards'									
$\lambda_1 = 2$*,	$\lambda_2 = 1$	$U(0, 1)$†	20	.163	.337	.140	.363	.174	.401
			50	.436	.669	.426	.677	.497	.748
		$U(0, 2)$	20	.236	.437	.183	.428	.244	.486
			50	.560	.772	.550	.782	.654	.842
Fig. 3b. 'Late difference'									
$W(2, 2)$‡	$W(.5, .5)$	$U(0, 1)$	20	.259	.444	.014	.058	.098	.252
			50	.727	.870	.016	.047	.256	.534
		$U(0, 2)$	20	.675	.832	.068	.183	.322	.605
			50	.994	.999	.104	.282	.830	.966
Fig. 3c. 'Late difference'									
$\lambda_1 = 1$,	$\lambda_2 = 1: t \in (0, .4)$	$U(0, 1)$	20	.118	.269	.027	.116	.078	.250
$\lambda_1 = 2$,	$\lambda_2 = .04: t \in (.4, \infty)$		50	.372	.583	.046	.158	.217	.443
		$U(0, 2)$	20	.612	.793	.134	.305	.399	.658
			50	.979	.995	.284	.536	.838	.952
Fig. 3d. 'Middle difference'									
$\lambda_1 = 2$,	$\lambda_2 = 2: t \in (0, .1)$	$U(0, 1)$	20	.238	.448	.122	.306	.099	.258
$\lambda_1 = 3$,	$\lambda_2 = .5: t \in (.1, .4)$		50	.681	.848	.322	.594	.247	.514
$\lambda_1 = .5$,	$\lambda_2 = 3: t \in (.4, .7)$	$U(0, 2)$	20	.257	.481	.095	.308	.065	.198
$\lambda_1 = 1$,	$\lambda_2 = 1: t \in (.7, \infty)$		50	.740	.892	.303	.548	.156	.341
Fig. 3e. 'Early difference'									
$\lambda_1 = 3$,	$\lambda_2 = .5: t \in (0, .2)$	$U(0, 1)$	20	.190	.418	.171	.422	.074	.206
$\lambda_1 = .5$,	$\lambda_2 = 3: t \in (.2, .4)$		50	.737	.915	.512	.785	.138	.351
$\lambda_1 = 1$,	$\lambda_2 = 1: t \in (.4, \infty)$	$U(0, 2)$	20	.218	.451	.147	.345	.048	.174
			50	.763	.921	.379	.638	.078	.257

* λ_i denotes the constant hazard function of S_i, a piecewise exponential distribution.
† $U(0, k)$ denotes a uniform distribution over the interval $(0, k)$.
‡ $W(\lambda, \alpha)$ denotes a Weibull distribution having survival function $S_i(t) = \exp\{-(\lambda t)^\alpha\}$.

one-sided test procedure in lightly censored, moderately censored and heavily censored data, specifically in data 0%, 25%, 43%, 63% or 79% censored. Note that the generalized Smirnov procedure tends to be conservative in small samples of heavily censored data.

Table 5 contains the results obtained from Monte Carlo simulations employed to evaluate the power of the generalized Smirnov test procedure in censored data. Graphs of the configurations corresponding to those in this table are shown in Fig. 3. When attempting to detect alternatives to H_0 in which large survival differences are most evident later in time, one would expect the logrank test procedure to be preferable to the Gehan–Wilcoxon. In our simulations of configurations of this type, which were non-Lehmann alternatives (see Figs 3b, 3c), the generalized Smirnov procedure was clearly more powerful than the logrank procedure, with the power of the Gehan–Wilcoxon test

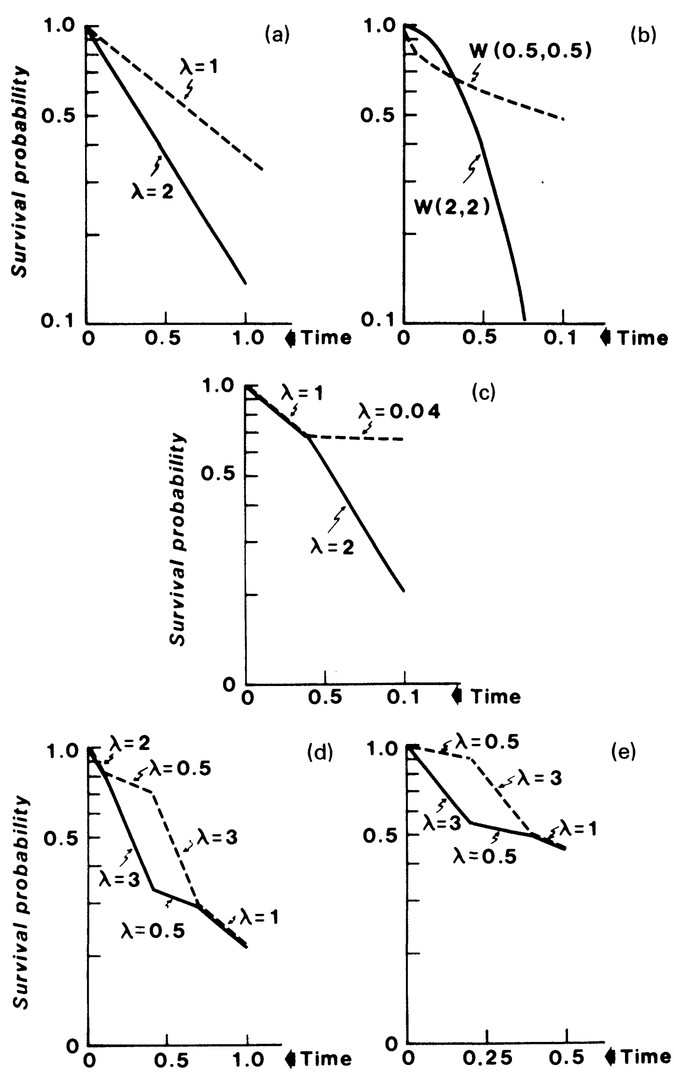

Figure 3. Semilog plots of survival distributions simulated for evaluation of power in censored data. Survival distributions are either piecewise exponentials with constant hazard over intervals or are Weibull distributions $W(\lambda, \alpha)$ having survival function $S(t) = \exp\{-(\lambda t)^{\alpha}\}$.

unacceptably low. When considering Lehmann alternatives (see Fig. 3a), the logrank procedure is more powerful, as one would expect, than either the generalized Smirnov or Gehan–Wilcoxon procedures. On the other hand, when attempting to detect alternatives to H_0 in which large survival differences are most evident earlier in time, one would expect the Gehan–Wilcoxon test procedure to be more sensitive than the logrank. In our simulations of configurations of this type (see Figs 3d, 3e) the power of the generalized Smirnov procedure was markedly better than that of the Gehan–Wilcoxon procedure, with the logrank test now having unacceptably low sensitivity.

5. Discussion

The classical Smirnov two-sample test procedure is sensitive to differences between two survival distributions, which are large at some point in time, whether or not differences exist at other time points. This fact, together with our experience that substantial differences are frequently most apparent at only one point in time, led us to surmise that a generalized Smirnov procedure would often be more sensitive than existing procedures for the type of data encountered in actual practice. The results of the Monte Carlo simulations suggest that, for the particular nonproportional hazards configurations inspected, the generalized Smirnov procedure is, in fact, more powerful than either the Gehan–Wilcoxon or the logrank test procedures.

However, it should be emphasized that these results certainly do not imply in general a superiority of the generalized Smirnov procedure over the Gehan–Wilcoxon or logrank procedures. The latter two are classical procedures for censored data, each having been shown to be very powerful in their abilities to detect certain types of differences between survival distributions. Specifically, the Gehan–Wilcoxon test procedure, as demonstrated by Lee *et al.* (1975) and Prentice and Marek (1979), is very sensitive to large early differences between distributions, particularly in situations in which the ratio of the hazard function of the better distribution to that of the worse is increasing in time but is never greater than one. On the other hand, when the ratio of the hazard functions is nearly constant in time (i.e. a Lehmann alternative), it has been proved that the logrank statistic provides a fully efficient rank test procedure.

Our general conclusion, then, is that the Smirnov procedure proposed here should be viewed as complementing the Gehan–Wilcoxon and logrank procedures and not as a competitor. The procedure to test equality of two survival distributions when the data are subject to arbitrary right censorship will depend on the type of distributional differences for which one desires particular sensitivity. The generalized Smirnov procedure will be appropriate when one wishes to have sensitivity to differences which are large at some point in time, whatever differences exist elsewhere. In fact, the sensitivity of the generalized Smirnov procedure to alternatives as general as this might make it the procedure of choice if one entered an experiment without a preconceived notion as to the specific type of alternative expected.

Acknowledgements

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Résumé

Le test classique de Kolmogorov de bon ajustement d'un échantillon, et celui de Smirnov, pour deux échantillons, sont modifiés pour que leur puissance augmente lorsq'on les applique à des données censurées. Ces deux tests sont généralisés pour traiter des données arbitrairement censurées à droite. Le concept de base sur lequel repose la procédure pour des données non censurées est repris de façon heuristique pour amener la formulation de cette procédure généralisée; des résultats de convergence asymptotique faible fournissent un support théorique. La taille et la puissance de la procédure généralisée de Smirnov pour deux échantillons, sont évaluées à l'aide de simulation de Monte-Carlo avec des échantillons de petite et de moyenne taille. Les résultants de simulation sont également utilisés pour faire des comparaisons avec les tests de Gehan–Wilcoxon et du log des rangs à deux échantillons: le test généralisé de Smirnov maintient la taille du test dans presque tous les cas étudiés, bien qu'il soit conservateur pour de petits échantillons de données fortement censurées. Il est plus puissant que le test de Gehan–Wilcoxon et que celui de log des rangs pour les alternatives non Lehmanniennes considérées.

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Appendix 1

Asymptotic Distribution Results

The following results provide the theoretical derivation of the asymptotic distribution results necessary in the formulation of generalized Kolmogorov goodness-of-fit and Smirnov two-sample test procedures proposed in this paper. The term 'asymptotic' in this paper refers to the limit as $N \to \infty$, or N_1 and $N_2 \to \infty$ with $N_1/N_2 \to \lambda$, a positive constant.

Hereafter let τ be an arbitrary positive number such that $\mathrm{pr}(Z_{ij} \geq \tau) > 0$, $i = 1, 2$, where $Z_{ij} = \min(X_{ij}, Y_{ij})$. 'Weak convergence,' denoted by $\Rightarrow$, will be used with respect to the Skorohod metric on the space $D(0, \tau)$ of functions on the interval $[0, \tau]$, which have discontinuities of only the first kind (Billingsley, 1968, Chapter 3).

Throughout this paper we assume that the cumulative hazard function of any survival distribution has a piecewise continuous derivative. Further, we assume that the distributional relationship between X_{ij} and Y_{ij} satisfies the following two conditions:

$$\frac{-\frac{\partial}{\partial t_1}\mathrm{pr}(X_{ij} \geq t_1, Y_{ij} \geq t_2)\Big|_{t_1=t_2=t}}{\mathrm{pr}(Z_{ij} \geq t)} = \frac{-\frac{d}{dt}S_i(t)}{S_i(t)}, \quad i = 1, 2, \tag{A.1}$$

whenever $\mathrm{pr}(Z_{ij} \geq t) > 0$, and

$$\mathrm{pr}(Z_{ij} \geq t) = S_i(t)C_i(t), \quad i = 1, 2. \tag{A.2}$$

Conditions (A.1) and (A.2) are both satisfied if X_{ij} and Y_{ij} are statistically independent.

Recall that our goodness-of-fit test procedures in censored data will be based upon inspection of the difference

$$Y_N(t) = \tfrac{1}{2}\{\hat{S}(t) + S^0(t)\} \int_0^t \{N\hat{C}(s^-)\}^{\frac{1}{2}} I_{[N(s)>0]}\, d\{\beta^0(s) - \hat{\beta}(s)\}.$$

The following lemma contains an important asymptotic distribution result needed in the formulation of the one-sample test procedures.

Lemma A.1. When H_0 is true, $\{Y_N(t): 0 \leq t \leq \tau\} \Rightarrow \mathcal{W}_s = \{\mathcal{W}_s(t): 0 \leq t \leq \tau\}$, where $\mathcal{W}_s$ is a time-transformed Brownian bridge. Specifically, if $W = \{W(t): t \geq 0\}$ represents Brownian motion, then $\mathcal{W}_s(t) = W\{1 - S(t)\} - \{1 - S(t)\}W(1)$. Note that for $0 \leq u \leq t \leq \tau$, $\mathrm{cov}\{\mathcal{W}_s(u), \mathcal{W}_s(t)\} = \{1 - S(u)\}S(t)$.

Proof. It follows directly from results in Fleming and Harrington (1981) that when the null hypothesis, $H_0: S(t) = S^0(t)$ for all t, holds,

$$\left[\int_0^t \{N\hat{C}(s^-)\}^{\frac{1}{2}} I_{[N(s)>0]}\, d\{\beta^0(s) - \hat{\beta}(s)\}: 0 \leq t \leq \tau\right] \Rightarrow Z \equiv \{Z(t): 0 \leq t \leq \tau\},$$

where Z is a mean zero Gaussian process possessing continuous sample paths, independent increments, and variance function

$$\mathrm{var}\, Z(t) = \{S(t)\}^{-1} - 1.$$

Aalen (1976) has proved that $\hat{\beta}(t)$ is a uniformly strongly consistent estimator of $\beta(t)$ over the interval $[0, \tau]$, from which it is straightforward to show that $\hat{S}(t)$ possesses the same property with respect to $S(t)$. Hence, from Corollary 1 of Theorem 5.1 in Billingsley (1968), it follows that $\{Y_N(t): 0 \leq t \leq \tau\} \Rightarrow \{S(t)Z(t): 0 \leq t \leq \tau\}$.

The proof of the lemma is completed by observing that, for $0 \leq s \leq t \leq \tau$,

$$\begin{aligned}\mathrm{cov}\{S(s)Z(s), S(t)Z(t)\} &= S(s)S(t)\,\mathrm{var}\, Z(s) \\ &= S(s)S(t)[\{S(s)\}^{-1} - 1] = S(t)\{1 - S(s)\}.\end{aligned}$$

Schey (1977), in generalizing the one-sided Kolmogorov goodness-of-fit test to the special case in which all observations are censored at τ, provided the following lemma which we too will use.

Lemma A.2. If $\mathcal{W}$ is a Brownian bridge over $[0, 1]$ and $x \in (0, 1)$, then

$$\mathrm{pr}\left(\sup_{0<t<x} \mathcal{W}(t) \geq y\right) = p(y, x),$$

where

$$p(y, x) = 1 - \Phi\{y/(x-x^2)^{\frac{1}{2}}\} + \Phi\{y(2x-1)/(x-x^2)^{\frac{1}{2}}\} \exp(-2y^2)$$

with Φ the standard normal cdf.

The following theorem, which contains the asymptotic distribution of the one- and two-sided Kolmogorov goodness-of-fit test statistics $\mathcal{Y}_N(\tau)$ and $\tilde{\mathcal{Y}}_N(\tau)$, is obtained directly by combining results of Lemmas A.1 and A.2.

Theorem A.1. For large N, when the null hypothesis is true,

$$\text{pr}\left\{\sup_{0\leqslant t\leqslant \tau} Y_N(t) \geqslant y\right\} \simeq p\{y, 1-S(\tau)\}.$$

When $p\{y, 1-S(\tau)\} \leqslant .40$,

$$\text{pr}\left\{\sup_{0\leqslant t\leqslant \tau} |\,Y_N(t)| \geqslant y\right\} \simeq 2p\{y, 1-S(\tau)\}.$$

Recall now that we will base our two-sample test procedure in censored data upon inspection of the difference

$$Y_{N_1,N_2}(t) = \tfrac{1}{2}\{\hat{S}_1(t)+\hat{S}_2(t)\}\int_0^t \left\{\frac{N_1\hat{C}_1(s^-)N_2\hat{C}_2(s^-)}{N_1\hat{C}_1(s^-)+N_2\hat{C}_2(s^-)}\right\}^{\frac{1}{2}} I_{[N_1(s)N_2(s)>0]}\, d\{\hat{\beta}_1(s)-\hat{\beta}_2(s)\}.$$

The following lemma contains the key asymptotic distribution result needed in the formulation of the two-sample test procedures.

Lemma A.3. As N_1, $N_2 \to \infty$, when the null hypothesis is true,

$$\{Y_{N_1,N_2}(t)\colon 0 \leqslant t \leqslant \tau\} \Rightarrow \mathcal{W}_s.$$

Proof. It again follows directly from results in Fleming and Harrington (1981) that when the null hypothesis, $H_0: S_1(t) = S_2(t)$ for all t, holds,

$$\left[\int_0^t \left\{\frac{N_1\hat{C}_1(s^-)N_2\hat{C}_2(s^-)}{N_1\hat{C}_1(s^-)+N_2\hat{C}_2(s^-)}\right\}^{\frac{1}{2}} I_{[N_1(s)N_2(s)>0]}\, d\{\hat{\beta}_1(s)-\hat{\beta}_2(s)\}\colon 0\leqslant t\leqslant \tau\right] \Rightarrow Z.$$

The remainder of the proof follows identically to that given for Lemma A.1.

The following theorem, which contains the asymptotic distribution of the one- and two-sided Smirnov two-sample test statistics $\mathcal{Y}_{N_1,N_2}(\tau)$ and $\tilde{\mathcal{Y}}_{N_1,N_2}(\tau)$, is proved directly by combining results of Lemmas A.2 and A.3.

Theorem A.2. Let $H_0: S_1(t) = S_2(t)\,[=S(t)$ unspecified$]$ for all t. For large N_1, N_2, when H_0 is true,

$$\text{pr}\left\{\sup_{0\leqslant t\leqslant \tau} Y_{N_1,N_2}(t) \geqslant y\right\} \simeq p\{y, 1-S(\tau)\}.$$

When $p(y, 1-S(\tau)) \leqslant .40$,

$$\text{pr}\left\{\sup_{0\leqslant t\leqslant \tau} |Y_{N_1,N_2}(t)| \geqslant y\right\} \simeq 2p\{y, 1-S(\tau)\}.$$

Appendix 2

Proof of Lemma 1

Assume without loss of generality that $\beta_l(t) > \beta_u(t)$, and set $\beta_l(t) = \beta_u(t)+x$, $x>0$. Then it suffices to show that

$$\tfrac{1}{2}\{S_l(t)+S_u(t)\}\{\beta_l(t)-\beta_u(t)\} - \{S_u(t)-S_l(t)\}$$
$$= \tfrac{1}{2}\exp\{-\beta_u(t)\}[\{\exp(-x)+1\}x - 2\{1-\exp(-x)\}] > 0.$$

This in turn holds if

$$f(x) = \{\exp(-x)+1\}x - 2\{1-\exp(-x)\} > 0 \text{ for } x > 0.$$

Thus the proof is completed by the facts that $f(0)=0$, $f'(0)=0$, and $f''(x) = x\exp(-x) > 0$ for $x>0$.